

Mean field theory of spin liquid states and topological orders*

X.G. WEN

Department of Physics
MIT
77 Massachusetts Avenue
Cambridge, MA 02139

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In this article we will review some recent developments in the mean field theory of spin liquid states. In particular we will emphasize on the importance of the gauge interaction in the mean field theory. We will demonstrate how to apply the mean field theory (with the gauge interaction) to study the physical properties of the spin liquids. The characterization of the spin liquids and the concept of the topological order are also discussed.

The article is arranged as the following. In section 1 we discuss the relation between spin liquid and high T_c superconductors. In section 2 we review the mean field approach to the spin liquids and discuss the effect of the gauge interactions in the mean field theory. In section 3 the chiral spin states and their effective theories are discussed. In section 4 and 5 we give the $SU(2)$ formalism and discuss the infrared stability of the mean field theory. In section 6 a new mean field state is studied which respect the time reversal symmetry and the parity. Such a mean field state correspond to the short ranged resonating-valence-bond (s-RVB) state conjectured before. In section 7 we discussed the ground states degeneracy of the spin liquids on compacted spaces with different topologies. We demonstrate that the chiral spin states and the s-RVB states contain new types of orderings which cannot be characterized by order parameters associated with broken symmetries. To describe such a new kind of orderings the concept of the topological order is introduced. We show that the ground state degeneracy is a universal property and can be used to characterize the universality classes of the topological orders in the spin liquid states. We summarize the article in section 8.

I. SPIN LIQUID STATES AND HIGH T_c SUPERCONDUCTORS

One of the most important properties of the high T_c superconductors is that the materials are insulators when the conduction band is *half* filled. The charge gap obtained by the optical conductance experiments is about $2eV$. Thus the high T_c materials are effectively Mott insulators.¹ The strong correlations are expect to play very important role in understanding high T_c superconductors. Within the one band Hubbard model description of the $Cu-O$ superconductors, the charge gap is due to the strong on site repulsion U . At half filling the only low lying excitations are spin fluctuations and the Hubbard model is reduce to a spin system. The energy scale of the spin fluctuations is the spin-spin coupling constant J . The experimental data suggest that $J \sim 0.1eV$.

There are many different approach to the high T_c superconductors. In the spin liquid approach, the strategy is to try to understand the properties of the high T_c superconductors from the low doping limit. In this approach we first study the spin liquid states at half filling. Then we try to understand the dynamics of a few doped holes in the spin liquid states and to obtain the properties of the high T_c superconductors at low doping.

Let us consider one hole in a spin state described by the t - J model. Since hole hopping is always accompanied with back flow of spins, the hole is always dressed with spin fluctuations. The spin-spin correlations in the spin state have a profound effect on the low energy dynamical properties of a hole. To understand the dynamics of a single doped hole we need to first understand the spin state at low energies. For example if the spin ground state has a ferromagnetic correlation the effective mass of the hole is of order $1/ta^2$. On the other hand if the spin ground state has a antiferromagnetic correlation the effective mass² of the hole will be of order $1/Ja^2$. Later we will see that the spin correlations can even change the statistics and quantum numbers of the hole. Depending on different spin

correlations, a hole may behave like a fermion, a boson or some times even an anyon. The hole may carry spin $1/2$ or zero spin for different situations.

Now let us discuss how to actually realize the low doping limit. The high T_c superconductors are in a spin disordered state at high doping. As we decrease the doping x , the material experience a phase transition from the spin disordered state to a antiferromagnetic state (at $x \sim 0.03$ for L_a-Cu-O). Because the spin correlations are different in the two phases, the low energy dynamics of the holes are expected to be different. Due to the phase transition we can not directly reach the low doping limit of the spin disordered state by simply reducing x . To obtain the low doping limit we need to add by hand some spin-spin interactions as we decrease the hole doping. The new spin-spin interactions are chosen to replace the frustration introduced by the hole motions and to qualitatively maintain the spin-spin correlations. In this case we may reach the low doping limit without encounter a phase transition.

We would like to remark that the frustration introduced by hole motions are dynamical in nature and are retarded. But the frustrations from the spin coupling are not retarded. So the two kinds of frustrations can never exactly replace each other. However, here we only require that the spin-frustrations are similar enough to the hole-frustrations that one can continuously reach the low doping limit without encounter the phase transition. The frustrations introduced by the hole hopping could be very complicated when represented by spin coupling. The hole hopping may even introduce multi-spin coupling.

In this article we are only interested in the spin liquid states that have a possibility to be continuously connected to the superconducting states. These spin liquid states should have the same symmetries and the same type of spin correlations as those in the doped high T_c samples. Thus the spin liquid states we are interested in are disordered (no long range spin-spin correlations) and respect the translation symmetry and the rotation symmetry. The spin liquid states that satisfy the above conditions can be divided into two classes: A) Gapless spin liquid states. An example of the gapless spin liquid state is the exact solution of one dimensional $S = \frac{1}{2}$ Heisenberg spin chain. B) Gapful spin liquid states. Examples are the spin-Peierls states and the chiral spin states. In this article I will concentrate on the spin liquid states with finite energy gaps.

The frustrated spin systems at more than one dimensions are very difficult to solve. Up to now we do not have any reliable approach to the problem. In this article we are going to review the mean field approach to the spin liquid states.^{3,4} Of course, the mean field theory of spin liquid states is not reliable due to quantum fluctuations. We can not use the mean field to solve a spin Hamiltonian and to determine, say, what kind of ground state is supported by that Hamiltonian. Failing to solve frustrated spin Hamiltonians we will limit ourself to simpler problems. First we would like to ask what are the possible spin liquid states which have the same type of the symmetries and spin correlations as the one in the superconducting state. Second we would like to know what are the experimental implications of different spin liquid states, hoping to experimentally determine which spin liquid state actually describes the high T_c superconductors. For these simpler problems, the mean field theory is useful. The mean field theory can provide some insights about *possible* spin liquid states. Although it cannot determine specifically which spin Hamiltonians that actually support the spin liquid, the mean field theory can provide some clues about what type of the interactions may favor the spin liquid under consideration. The mean field theory can also tell us the characteristic properties of those spin liquid states so that if they are discovered in experiments or numerical calculations we are able to recognize them.

One of main problems we are going to address in this article is the self consistency of the mean field theory. In order for the mean field theory to have any chance to describe

the real spin liquids, it must have certain stability in the infrared limits. In other words a good mean field ground state should be a generic state *i.e.*, be stable against any small perturbations. If such a stability exists, the mean field ground state may have some chance to survive the quantum fluctuations and to describe a real spin liquid (qualitatively). In this case the mean field theory describes a universality class of spin liquid states. Many mean field ground states studied before do not have this infrared stability. In this article we are going to outline sufficient conditions under which the mean field ground states are dynamically stable.

A closely related question is what are the quantum numbers that characterize the universality classes of spin liquid states. Certainly the universality classes may be characterized by order parameters associated broken symmetries. However some spin liquid states contain internal structures (or in other words, new types of orderings) which can not be characterized by local order parameters. Such new types of ordering are called topological orders which is another important subject we are going to review in this article.

II. MEAN FIELD THEORY OF THE SPIN LIQUID STATES

In this section I will briefly review the mean field approach to the spin liquid state developed in Ref. 3.

At half filling the Hubbard model reduces to the Heisenberg model:

$$H = \sum_{\langle ij \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j \quad (1)$$

To obtain the mean field ground state of the spin liquids we introduce the slave fermion operators $s_{i\alpha}$, $\alpha = 1, 2$ which carries spin $1/2$. The spin operator $\mathbf{S}_i$ is represented as

$$\mathbf{S}_i = s_{i\alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} s_{i\beta} \quad (2)$$

In terms of the slave fermion operators the Hamiltonian (1) can be rewritten as

$$H = \sum_{\langle ij \rangle} -2J_{ij} s_{i\alpha}^\dagger s_{j\alpha} s_{j\beta}^\dagger s_{i\beta} \quad (3)$$

Notice that the Hilbert space of (3) is larger than that of (1). The equivalence between (1) and (3) is valid only in the subspace where there is exactly one slave fermion per site. Therefore to use (3) to describe the spin state we need to impose the constraint

$$s_{i\alpha}^\dagger s_{i\alpha} = 1, \quad s_{i\alpha} s_{i\beta} \epsilon_{\alpha\beta} = 0 \quad (4)$$

The second constraint is actually a consequence of the first one.

A mean field ground state at “zeroth” order is obtained by making the following approximations. First we replace constraint (4) by its vacuum average

$$\langle s_{i\alpha}^\dagger s_{i\alpha} \rangle = 1 \quad (5)$$

Such a constraint can be enforced by including a *site dependent* and time independent Lagrangian multiplier: $a_0(i)(s_{i\alpha}^\dagger s_{i\alpha} - 1)$ in the Hamiltonian. At the zeroth order we ignore the fluctuations (*i.e.*, the time dependence) of a_0 . If we included the fluctuations of the a_0 the constraint (5) would become the original constraint (4).^{5,6} Second we replace the operator $s_{i\alpha}^\dagger s_{j\alpha}$ by its vacuum expectation value χ_{ij} , again ignoring their fluctuations. In this way we obtain the zeroth order mean field Hamiltonian:

$$H_{mean} = \sum_{i,j} -J_{ij} \left[(s_{i\alpha}^\dagger s_{j\alpha} \chi_{ji} + h.c) - |\chi_{ij}|^2 \right] + \sum_i a_0(i)(s_{i\alpha}^\dagger s_{i\alpha} - 1) \quad (6)$$

χ_{ij} in (6) must satisfy the self consistency condition

$$\chi_{ij} = \langle s_{i\alpha}^\dagger s_{j\alpha} \rangle \quad (7)$$

and the site dependent chemical potential $a_0(i)$ is chosen such that (5) is satisfied by the mean field ground state. For ease of speaking we will call χ_{ij} the “order parameters” of the spin liquid state. The fluctuations in χ_{ij} and $a_0(i)$ describe the collective excitations above the mean field ground state.

Let us discuss two mean field ground states, spin-Peierls state⁷ and flux state,⁸ in more detail. The spin-Peierls state is described by the following “order parameter”:

$$\begin{aligned} \chi_{i,i+\hat{x}} &= \frac{1}{2}(1 + (-)^{i_x}) \\ \text{others} &= 0 \end{aligned} \quad (8)$$

One can easily check the self consistency (7) is satisfied by the mean field ground states of (6). The slave fermion spectrum has no dispersion in the mean field spin-Peierls state:

$$E_k = \pm 2J_1 \quad (9)$$

where J_1 is the nearest neighbor spin coupling. The valence band with $E_k = -2J_1$ is completely filled by the slave fermions. The spin excitations have finite energy gap in the spin-Peierls state.

The flux state is given by the “order parameter”

$$\begin{aligned} \chi_{i,i+\hat{x}} &= i\eta_1 (-)^{i_x} \\ \chi_{i,i+\hat{y}} &= i\eta_1 \\ \text{others} &= 0 \end{aligned} \quad (10)$$

The mean field Hamiltonian (6) is equivalent to electron hopping problem with π flux per plaquette. The slave fermion spectrum in the flux phase is given by

$$E_k = \pm J_1 \eta_1 \sqrt{\sin(k_x a)^2 + \sin(k_y a)^2} \quad (11)$$

where $-\frac{\pi}{2a} < k_x < \frac{\pi}{2a}$ and $-\frac{\pi}{a} < k_y < \frac{\pi}{a}$. The fermi surfaces at half filling are points at $(k_x, k_y) = (0, 0)$ and $(0, \frac{\pi}{a})$. At mean field level the flux state contain gapless spin excitations which correspond to the particle-hole excitations across the Fermi “points”.

In addition to those gapless spin excitations the mean field flux state also contain other collective excitations at low energies. Such excitations are described by the fluctuations of a_0 and a_{ij} , where a_{ij} are the phases of χ_{ij} :

$$\chi_{ij} = \chi_{ij}^{(mean)} e^{ia_{ij}} \quad (12)$$

(We ignored the amplitude fluctuation of χ_{ij} because the amplitude fluctuations have finite energy gap and are not essential in our discussion.) Including those fluctuations the mean field Hamiltonian becomes

$$H = \sum_{i,j} -J_{ij} (s_{i\alpha}^\dagger s_{j\alpha} \chi_{ji}^{(mean)} e^{ia_{ji}} + h.c) + \sum_i a_0(i) (s_{i\alpha}^\dagger s_{i\alpha} - 1) \quad (13)$$

We would like to call (13) the first order mean field Hamiltonian. From (13) we see that fluctuations described by a_0 and a_{ij} are nothing but lattice gauge fields.⁵ The Hamiltonian is invariant under the gauge transformation

$$\begin{aligned} a_{ij} &\rightarrow a_{ij} + \theta_i - \theta_j \\ s_i &\rightarrow s_i e^{i\theta_i} \end{aligned} \quad (14)$$

Now let us consider the interactions between the spin excitations and the gauge fluctuations. For convenience let us add a Maxwell term $\frac{1}{8\pi g^2} (v^2 f_{12}^2 - f_{0i}^2)$ to Hamiltonian (13), where g is a coupling constant and $f_{\mu\nu}$ is the field strength of a_μ . The original theory correspond to $g \rightarrow \infty$ limit. The Maxwell term is generated in the process of the renormalization. At energy scale Λ , $g^{-2}(\Lambda) \sim \Lambda^{-1} - \Lambda_0^{-1}$ where Λ_0 is the energy cutoff in the original theory. Due to the coupling with the gauge field, a slave fermion creates a “electric” field f_{0i} of the gauge field (a_0, a_{ij}) (notice a slave fermion carries a unit charge of the gauge field). The potential energy between a pair of the particle and the hole excitations is

$$V(\mathbf{r}_1 - \mathbf{r}_2) = 2\pi g^2 \log |\mathbf{r}_1 - \mathbf{r}_2| \quad (15)$$

We find that the particle and the hole are confined since $V(r) \rightarrow \infty$ as $r \rightarrow \infty$.⁹ Because of the confinement the slave fermions cannot appear as the quasiparticles in the physical spin liquid state. The existence of the long range confining interaction indicates infrared divergence in the theory. This results suggest that the quantum fluctuations (*e.g.*, the gauge fluctuations) in the flux state are very important and may drastically change the low energy properties of the zeroth order mean field theory. In this case the zeroth order mean field theory does not provide any reliable information about the low energy properties of spin liquids.

Notice that at the “zeroth” order, the slave fermion density in the mean field flux phase does not satisfy the constraint (4). The slave fermion density fluctuations are even gapless. In this case, because the density fluctuation is so strong and the constraint (4) is so badly violated, the mean field theory (6) comes nowhere close to describe the low energy properties of a true spin liquid state (which contains no density fluctuations). To cure this problem we should include the gauge fluctuations and use the Hamiltonian (13). But the Hamiltonian describe a strongly interacting system of the slave fermions and the gauge bosons. It is difficult to find the ground state of (13). Solving (13) may not be easier then solving the original Heisenberg model. If we can not solve (13) we have to say the

mean field theory is not so useful in the sense that the mean field theory does not even provide qualitative informations about the low energy properties of spin liquid states.

But, there are exceptions to the above misfortune. In those cases the low energy properties of spin liquid states can be relatively reliably determined from the mean field theory.

III. CHIRAL SPIN LIQUID STATE

In this section we will discuss a situation where the mean field theory does tell us some things about the spin liquid state, at least qualitatively. The mean field order parameter χ_{ij} in this case generates flux. The slave fermions described by (6) behave as if they are moving in a magnetic field. When the flux have a correct commensuration with the slave fermion density (which is one fermion per site) and the lattice, an integral number of Landau levels (or more precisely, Landau bands due to the lattice) are completely filled. The slave fermion gas is then incompressible because of the finite gap between Landau levels. The slave fermion density (and the current) fluctuations are absent (at low energies) even at the “zeroth” order of the mean field theory. Therefore the mean theory in this case does provide a qualitative correct description of the spin liquid state. The time reversal symmetry and the parity are found to be spontaneously broken. The corresponding spin states are called chiral spin states.^{10,11,12,13}

After include the vacuum polarization of the slave fermions, the effective lattice gauge theory contains a Chern-Simons term which gives the gauge field fluctuations a finite energy gap. Thus the interaction mediated by the gauge field is short ranged and not confining. The slave fermion in the mean field theory will have a finite energy and appear in physical spectrum even after we include the gauge fluctuations. Such spin $\frac{1}{2}$ neutral quasiparticles will be called spinons. The Chern-Simons term also gives the spinons a fractional statistics.

In the following I will study a specific mean field chiral spin state to demonstrate the picture discussed above.¹¹ The mean field “order parameter” χ_{ij} for the chiral spin state is non zero on both the nearest neighbor links and the second neighbor links. χ_{ij} on the nearest neighbor links are give by (10). χ_{ij} on the second neighbor links are given by

$$\begin{aligned}\chi_{i,i+\hat{x}+\hat{y}} &= -i\eta_2(-)^{i_x} \\ \chi_{i,i+\hat{x}-\hat{y}} &= i\eta_2(-)^{i_x}\end{aligned}\tag{16}$$

We know χ_{ij} in (10) induce π flux per plaquette. χ_{ij} in (16) induce $\frac{1}{2}\pi$ flux for each triangle.

The π flux phase discussed in last section is some how special. Since π flux is equivalent to $-\pi$ flux, the π flux phase respect time reversal symmetry (T) and parity (P). As a consequence the energy gap is not opened between the conduction and the valence bands. However $\frac{1}{2}\pi$ flux phase described by (10) and (16) is not equivalent to $-\frac{1}{2}\pi$ flux phase. Under T or P , $\frac{1}{2}\pi$ flux change into $-\frac{1}{2}\pi$ flux and η_2 in (16) change into $-\eta_2$. Therefore the chiral spin state with non zero η_2 break T and P spontaneously. Using the identity

$$E_{123} \equiv \mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3) = 2i(\hat{\chi}_{12}\hat{\chi}_{23}\hat{\chi}_{31} - \hat{\chi}_{13}\hat{\chi}_{32}\hat{\chi}_{21}), \quad \hat{\chi}_{ij} = s_i^\dagger s_j \tag{17}$$

we find that $\langle E_{123} \rangle \sim \text{Im}(\chi_{12}\chi_{23}\chi_{31})$ is non zero in the chiral spin state. Notice that E_{123} is odd under T or P , thus E_{123} can be regarded as the T and P breaking order parameter.

Now let us discuss when the mean field Hamiltonian (6) support the chiral spin state. Let J_1 denote the nearest neighbor spin coupling and J_2 the second neighbor one. When $\eta_2 = 0$ the slave fermion spectrum is given by (11). The conduction band and the valence band touch at points $(k_x, k_y) = (0, 0), (0, \frac{\pi}{a})$. We find that a non zero η_2 open a energy gap between the conduction band and the valence band. The slave fermion spectrum is given by

$$E_k = \pm \sqrt{J_1^2 \eta_1^2 (\sin(k_x a)^2 + \sin(k_y a)^2) + J_2^2 \eta_2^2 (\cos(k_x a + k_y a) + \cos(k_x a - k_y a))^2} \quad (18)$$

Therefore in the chiral spin state the slave fermion excitations have a finite energy gap. The mean field ground state is obtained by filling the valence band. One can easily check that the spin-spin correlation is short ranged. Using the ground state wave function we can calculate $\langle s_{\alpha i}^\dagger s_{\alpha j} \rangle$ to check the self consistency condition (7). We find that when $J_2/J_1 < 0.49$ the equation (7) only support a solution with $\eta_1 \neq 0$ and $\eta_2 = 0$, *i.e.*, the π flux phase. When $J_2/J_1 > 0.49$ the equation (7) also support a second solution with $\eta_2 \neq 0$. The second solution is found to have lower mean field energy. In this case T and P are spontaneously broken.

We would like to stress that the mean field calculation are not reliable quantitatively due to the strong quantum fluctuations. The above calculations are just for illustrative purpose. To really determine whether a Hamiltonian supports chiral spin state or not, one has to rely on exact numerical calculations. In the following we will assume the existence of the chiral spin state and discuss its consequences.

Before study the properties of the quasiparticles in the chiral spin state we need to first discuss a peculiar property of the mean field ground state. Notice that the mean Hamiltonian for the chiral state is equivalent to a problem of electron hopping in magnetic field. The coupling between the slave fermions and the gauge field a_μ is identical to the coupling between electrons and the electromagnetic field. Thus one expects that the slave fermion system described by (13) has a phenomenon similar to the Hall effect. The “Hall effect” in this case implies that an “electric” field of the a_μ gauge field induces a slave fermion current in the transverse direction:

$$j_x = \sigma_{xy} e_y, \quad e_y = \partial_t a_y - \partial_y a_0 \quad (19)$$

where σ_{xy} is the Hall conductance. There is a theorem stating that the Hall conductance of a fill band is always quantized as an integer times $1/2\pi$.¹⁴ In our case this implies $\sigma_{xy} = 2n/2\pi$, where the factor 2 comes from the spins. To obtain the value of the “Hall” conductance we need to calculate the slave fermion current correlation function which has the following form for small $\omega(=k_0)$ and k_i :

$$K_{\mu\nu} \equiv \langle j_\mu(k) j_\nu(-k) \rangle = \sigma_{xy} k_\lambda \epsilon_{\mu\nu\lambda} + \dots \quad (20)$$

The above current correlation function has been discussed in detail in Ref. 14,11. We find in our case

$$\sigma_{xy} = 2/2\pi. \quad (21)$$

The simplest way to understand this result is to realize that after turning off the lattice potential, the valence band in the mean field chiral spin state becomes the first Landau

level. Both the spin up and spin down slave fermions in the valence band contribute $1/2\pi$ to the “Hall” conductance. Thus the total “Hall” conductance is given by (21).

The effective action for the gauge fluctuations is obtained by integrating out the slave fermions in (13). Using the current correlation function $K_{\mu\nu}$ we can easily write down the effective action in the continuum limit:

$$\begin{aligned} S &= \int d^3x d^3x' \frac{1}{2} a_\mu(x) K_{\mu\nu}(x-x') a_\nu(x') \\ &= \int d^3x \frac{1}{2} \sigma_{xy} a_\mu \partial_\nu a_\lambda \epsilon_{\mu\nu\lambda} + \frac{1}{8\pi g^2} (\mathbf{e}^2 - v^2 b^2) + \dots \end{aligned} \quad (22)$$

where $e_i = \partial_0 a_i - \partial_i a_0|_{i=x,y}$ is the “electric” field and $b = \partial_x a_y - \partial_y a_x$ the “magnetic” field of the gauge field a_μ . g^2 in (22) is of order the slave fermion gap and v is of order $1/aJ$ the typical slave fermion velocity. Using the effective Lagrangian we can calculate the low energy dynamical properties of the gauge fluctuations. In section 7 we will use (22) to calculate the ground state degeneracy of the chiral spin state on torus.

Due to the non zero “Hall” conductance σ_{xy} the mean field chiral spin state has a peculiar property that we can change the slave fermion density *without* creating slave fermions in the conduction band or creating holes in the valence band. This property is important in understanding the properties of the quasiparticles. Let us slowly turn on the flux of the a_μ field, $\Phi = \int d^2x b$. If the flux is distributed in a large region, the “magnetic” field b in the flux is small. In this case the energy gap between the valence band and the conduction band remain to be finite and all the energy levels in the valence band are filled by a spin up and a spin down slave fermions. Turning on the flux induces a circular “electric” field e_θ which in turn generates a slave fermion current in the radial direction $\hat{\mathbf{r}}$ due to the non zero σ_{xy} . Thus some charges are accumulated near the origin. We can easily find that the total number of the induced slave fermions is given by

$$N \equiv \sum_i (\langle s_{\alpha i}^\dagger s_{\alpha i} \rangle - 1) = -\sigma_{xy} \Phi = -\frac{\Phi}{\pi} \quad (23)$$

We would like to stress that the flux only change the slave fermion density. It does not induce any spin quantum number. No matter how many slave fermions are induced by the flux, the flux tube always carries zero spin because every energy levels in the valence band is filled by a spin up and a spin down electron.

Now we are ready to discuss the quantum number of the quasiparticle excitations in the chiral spin state. The simplest excitation in the mean field chiral spin state can be obtained by adding a slave fermion in the conduction band. However this excitation is not physical because the additional slave fermion in the conduction band violate the constraint (5). To satisfy the constraint we can add flux to change the slave fermion density in the valence band. From the above discussion we see that the extra slave fermion density arisen from the slave fermion in the conduction band can be canceled by introducing π flux (see (23)). Therefore the physical quasiparticles—the spinons—in the chiral spin state are slave fermions *dressed* by π flux. The spinons (*i.e.*, the dressed slave fermions) carry spin $\frac{1}{2}$ since the flux cannot induce any spin quantum numbers. However as a bound state of a charge and flux (note the slave fermion carry a unit charge of the a_μ gauge field), the spinon has a fractional statistics.¹⁵ Exchanging two spinons is equivalent to move one spinon half way around the other which induces a phase $\theta = \frac{1}{2}q\Phi$. Here $q = 1$ is the charge of the slave

fermion and $\Phi = \pi$ is the flux bounded to the slave fermion. Hence the statistical angle of the spinon is $\theta = \frac{\pi}{2}$. Particles with such a statistics is half way between bosons (with $\theta = 0$) and fermions (with $\theta = \pi$) and are called semions.

To understand the low energy dynamics of the spinons we would like to derive the effective Lagrangian for the spinons. First let us ignore the gauge fluctuations by setting $a_\mu = 0$. In this case a slave fermion in the conduction band and the valence band has the follow dispersion relation:

$$E_k^\pm = \pm \sqrt{J_1^2 \eta_1^2 (\sin(k_x a)^2 + \sin(k_y a)^2) + J_2^2 \eta_2^2 (\cos(k_x a + k_y a) + \cos(k_x a - k_y a))^2} \quad (24)$$

E_k^+ (E_k^-) has two minimums (maximums) at $(0, 0)$ and $(0, \pi/a)$. Therefore in the continuum limit we have two species of slave fermions described by the follow effective Lagrangian

$$\mathcal{L} = \sum_{I=1,2} \left[-i s_{I\alpha}^\dagger \partial_t s_{I\alpha} + \frac{1}{2m_s} s_{I\alpha}^\dagger \partial_i^2 s_{I\alpha} - i \bar{s}_{I\alpha}^\dagger \partial_t \bar{s}_{I\alpha} + \frac{1}{2m_s} \bar{s}_{I\alpha}^\dagger \partial_i^2 \bar{s}_{I\alpha} \right] \quad (25)$$

$s_{1\alpha}$ and $s_{2\alpha}$ correspond to the slave fermions near the two minimums of the conduction band and $\bar{s}_{1\alpha}$ and $\bar{s}_{2\alpha}$ correspond to the holes near the maximums of the valence band. When $a_\mu \neq 0$, the coupling between the slave fermions and the gauge field can be obtained by replacing ∂_μ in (25) by $\partial_\mu \pm i a_\mu$. Such form of the coupling is determined by the requirement of the gauge invariance. After including the coupling to the gauge field the total effective Lagrangian has a form

$$\begin{aligned} \mathcal{L} = & \sum_{I=1,2} \left[-i s_{I\alpha}^\dagger (\partial_t - i a_0) s_{I\alpha} + \frac{1}{2m_s} s_{I\alpha}^\dagger (\partial_i - i a_i)^2 s_{I\alpha} \right] \\ & + \sum_{I=1,2} \left[-i \bar{s}_{I\alpha}^\dagger (\partial_t + i a_0) \bar{s}_{I\alpha} + \frac{1}{2m_s} \bar{s}_{I\alpha}^\dagger (\partial_i + i a_i)^2 \bar{s}_{I\alpha} \right] \\ & - \frac{2}{4\pi} a_\mu \partial_\nu a_\lambda \epsilon_{\mu\nu\lambda} + \frac{1}{8\pi g^2} (\mathbf{e}^2 - v^2 b^2) \end{aligned} \quad (26)$$

From the equation of motion $\frac{\delta \mathcal{L}}{\delta a_0} = 0$ we find that

$$n_1 + n_2 = -\frac{1}{\pi} b \quad (27)$$

where $n_I = \sum_\alpha (s_{I\alpha}^\dagger s_{I\alpha} - \bar{s}_{I\alpha}^\dagger \bar{s}_{I\alpha})$, $I = 1, 2$ is the density of the slave fermions. From (27) we see that a slave fermion is dressed by π flux which agrees with the previous results. The statistics of the spinons can also be directly calculated from (26).¹⁵

We would like to point out that according to (27) there are four kind of quasiparticles for each fixed I : the spin up and down slave fermions in the conduction band and the spin up and down holes in the valence band. This results is *not* correct. For each fixed value of I there should only be two kinds of quasiparticles: the spin up and down spinons. The spin up slave fermions in the conduction band and the spin up hole in the valence band give rise to the same spinon after the Gutzwiller projection.⁶ This over counting problem can be resolved after we include the $SU(2)$ gauge symmetry (see section 4).

To study the properties of the doped holes, we must generalize the Heisenberg model (1) to the t - J model:

$$H = \sum_{\langle ij \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{\langle ij \rangle} t_{ij} c_i^\dagger c_j$$

$$c_i^\dagger c_i = 0, 1$$
(28)

Representing electrons in terms of the slave bosons b_i and the slave fermions $s_{i\alpha}$: $c_\alpha = s_\alpha b^\dagger$, we may write the t - J model as

$$H = \sum J_{ij} s_{i\alpha}^\dagger s_{j\alpha} s_{j\beta}^\dagger s_{j\beta} (1 - n_i)(1 - n_j)$$

$$+ \sum t s_{i\alpha}^\dagger s_{j\alpha} b_j^\dagger b_i$$
(29)

with the constraint:

$$s_{i\alpha}^\dagger s_{i\alpha} + b_i^\dagger b_i = 1$$
(30)

on the state in the physical Hilbert space. In (29) $n_i = b_i^\dagger b_i$ is the number of the holes on the site i . The mean field Hamiltonian at non-zero doping has a form

$$H_{mean} = \sum J_{ij} \chi_{ji} s_{i\alpha}^\dagger s_{j\alpha} + \sum t \chi_{ij} b_j^\dagger b_i + \sum [a_0 (s_{i\alpha}^\dagger s_{i\alpha} + n_i - 1) + \mu n_i]$$
(31)

At the zeroth order the fluctuations of χ_{ij} are ignored and the holon b has the following spectrum

$$E_k^h = -t\eta_1 \sqrt{\sin(k_x a)^2 + \sin(k_y a)^2} + a_0 + \mu$$
(32)

assuming (29) contains only nearest neighbor hole hopping. Because χ_{ij} is of order $O(1)$, the band width of the holon is of order t . The holon is very mobile in the chiral spin liquid state. This result suggests that the chiral spin liquid state studied here may be stabilized by the doping.

To obtain the quantum numbers for the doped holes, let us add one hole to the system. From (30) we see that the slave fermion number must be change by 1 near the hole in order to satisfy the constraint. There are two way to change the slave fermion density. A) We simply remove a slave fermion from the valence band. In this case the quasiparticle is a bound state of a slave fermion s_α and a hole b . The particle carries charge e spin $1/2$ and has a fermionic statistics. Such a quasiparticle is just the original electron. B) We may change the slave fermion density by turning on flux in a_μ field. In this case the quasiparticle is a bound state of π flux and a hole b . The quasiparticle – called holon – carries charge e spin 0 and has a fractional statistics $\theta = \pi/2$.

Following the discussion which leads to the spinon effective theory (26), we can obtain the effective theory for holons:

$$\mathcal{L} = \sum_{I=1,2} \left[-ib_I^\dagger (\partial_t + ia_0 - ieA_0) b_I + \frac{1}{2m_h} b_I^\dagger (\partial_i + ia_i - ieA_i)^2 b_I \right]$$

$$- \frac{2}{4\pi} a_\mu \partial_\nu a_\lambda \epsilon_{\mu\nu\lambda} + \frac{1}{8\pi g^2} (\mathbf{e}^2 - v^2 b^2)$$
(33)

Because (32) has two minimums, we have two species of holons in the continuum limits. The low energy dynamical properties of the holons can be calculated from (33).

From (33) we see that the holons in the chiral spin state are charged semions. At low temperatures a gas of the charged semions are shown to form a superconducting state.¹⁶ The transition temperature is of order n/m^* , where n and m^* are the density and the mass of the semions. The flux quantum in the superconducting state is found to be $hc/2e$ in agreement with experiments. It is very interesting to see whether the actual high T_c superconductors are described by the chiral spin state or not.

As an application of the effective theory let us consider the interaction of the holes with an external electromagnetic field. It was suggested that when polarized light is reflected from the T and P violating doped chiral spin state, its polarization will be rotated.¹⁷ This is a practical way to detect the T and P breaking in the high T_c superconductors. The relevant physics may be described by a bag picture in which the holon is viewed as a bound state of the charged hole and a “flux bag” (the “flux bag” is the region where $b \neq 0$). The size of the bag is of order v/g^2 . An external oscillating electric field drives the hole moving inside the bag which create a relative motion between the hole and the flux bag. The motion of the hole resembles the motion of a charged particle in a constant magnetic field which causes a rotation of the polarization of the light. We would like to emphasize that it is the relative motion between the hole and the flux bag which gives rise to observable T and P breaking effects. If the frequency of the oscillating electric field is low, the hole and the bag will move together so that there is no rotation in the polarization. After a calculation we find that the rotation angle of the polarization is given by $\phi = \frac{3}{4} \frac{\omega^2}{\omega_p m^* v^2}$ (for ω less than the spinon gap). Here ω_p is the plasma frequency, m^* is the effective mass of the holon and v is the velocity of the spin fluctuations. Several experiments have been done to test this prediction.¹⁸ But no consistent conclusion can be drawn from the experiments. Other phenomenological implications in the chiral spin states can be found in Ref. 19.

IV. $SU(2)$ FORMALISM

In the next section we are going to study another spin liquid state which can also be reliably described by the mean field theory. To study that spin liquid state we first need to generalize the mean field formalism discussed in section 2 and 3. The generalized mean field theory contain two “order parameters”

$$\begin{aligned}\eta_{ij} &= \epsilon^{\alpha\beta} s_{i\alpha} s_{j\beta} = \eta_{ji} \\ \chi_{ij} &= s_{i\alpha}^\dagger s_{j\alpha} = \chi_{ji}^\dagger\end{aligned}\tag{34}$$

The mean field Hamiltonian of (3) has a form

$$\begin{aligned}H_{mean} &= \sum J_{ij} \left[|\eta_{ij}|^2 + |\chi_{ij}|^2 - (\chi_{ji} s_{i\alpha}^\dagger s_{j\alpha} + \eta_{ij} s_{i\alpha} s_{j\beta} \epsilon^{\alpha\beta} + h.c.) \right] \\ &+ \sum \left[a_0^3 (s_{i\alpha}^\dagger s_{i\alpha} - 1) + [(a_0^1 + i a_0^2) s_{i\alpha} s_{i\beta} \epsilon_{\alpha\beta} + h.c.] \right]\end{aligned}\tag{35}$$

a_0^l are the Lagrangian multipliers to enforce the constraint (4).

The Hamiltonian (35) and the constraints (4) have a local $SU(2)$ symmetry.⁶ The local $SU(2)$ symmetry becomes explicit if we introduce

$$(\psi_{a\alpha}) = \begin{pmatrix} s_1 & s_2 \\ s_2^\dagger & -s_1^\dagger \end{pmatrix} \quad (36)$$

$$8 U_{ij} = \begin{pmatrix} -\chi_{ij}^\dagger & \eta_{ij} \\ \eta_{ij}^\dagger & \chi_{ij} \end{pmatrix} \quad (37)$$

Using (36) and (37) we can rewrite (35) as:

$$H_{mean} = \sum J_{ij} \text{Tr} \left[8 U_{ij}^\dagger U_{ij} + (\psi_i^\dagger U_{ij} \psi_j + h.c.) \right] + \sum a_0^l \text{Tr}(\psi_i^\dagger \tau^l \psi_i) \quad (38)$$

where τ^l , $l = 1, 2, 3$ are the Pauli matrices. From (38) we can see clearly that the Hamiltonian is invariant under a local $SU(2)$ gauge transformation W_i :

$$\begin{aligned} \psi_i &\rightarrow W_i \psi_i \\ U_{ij} &\rightarrow W_i U_{ij} W_j^\dagger \end{aligned} \quad (39)$$

A mean field ground state at “zeroth” order is again obtained by making the following approximations. First we replace constraint (4) by its vacuum average (5). Such a constraint can be realized by the *site dependent* Lagrangian multipliers $a_0^l(i)$ in the Hamiltonian. Second we assign specific values to η_{ij} and χ_{ij} and ignoring their quantum fluctuations. In this approximation the dynamics of the slave fermions is governed by the mean field Hamiltonian

$$H_{mean} = \sum J_{ij} \text{Tr}(\psi_i^\dagger U_{ij}^{(mean)} \psi_j + h.c.) + \sum a_0^l \text{Tr}(\psi_i^\dagger \tau^l \psi_i) \quad (40)$$

where $U_{ij}^{(mean)}$ is the mean field solution, which satisfies the self consistency condition

$$\chi_{ij}^{(mean)} = \langle s_{i\alpha}^\dagger s_{j\alpha} \rangle, \quad \eta_{ij}^{(mean)} = \langle s_{i\alpha} s_{i\beta} \epsilon_{\alpha\beta} \rangle \quad (41)$$

a_0^l in (40) are chosen such that (5) is satisfied.

The important fluctuations around the mean field ground state are the “phase” fluctuations of U_{ij} :

$$U_{ij} = U_{ij}^{(mean)} e^{iA_{ij}} \quad (42)$$

where $A_{ij} = a_{ij}^l \tau^l$ is a 2×2 traceless hermitian matrix. Due to the $SU(2)$ gauge symmetry those fluctuations may be gapless. We must include those phase fluctuations in order to obtain qualitatively correct results for spin liquid states at low energies. This lead us to the “first” order mean field theory. At the first order, the “phase” fluctuations of U_{ij} (i.e., A_{ij}) and the fluctuations of the Lagrangian multipliers a_0^l describe a $SU(2)$ lattice gauge field.^{5,6} It was shown that the $SU(2)$ gauge field fluctuations enforce the operator constraint (4). At the first order, the slave fermion dynamics is governed by:

$$H_{mean} = \sum J_{ij} \text{Tr}(\psi_i^\dagger U_{ij}^{(mean)} e^{ia_{ij}^l \tau^l} \psi_j + h.c.) + \sum a_0^l \text{Tr}(\psi_i^\dagger \tau^l \psi_i) \quad (43)$$

The above discussions can also be generalized to the t - J model (28). To have manifest $SU(2)$ invariant theory we need to introduce two slave bosons $b_a|_{a=1,2}$ which forms a $SU(2)$ doublet. Now the electron operator is represented as

$$c_\alpha = b_a^\dagger \psi_{a\alpha} \quad (44)$$

and (29) becomes

$$H = \sum J_{ij} s_{i\alpha}^\dagger s_{j\alpha} s_{j\beta}^\dagger s_{i\beta} (1 - n_i)(1 - n_j) \\ + \sum t b_{a,j}^\dagger \psi_{a\alpha,j} \psi_{b\alpha,i}^\dagger b_{b,i} \quad (45)$$

where $n_i = b_{1i}^\dagger b_{1i} + b_{2i}^\dagger b_{2i} = 0, 1$ is the hole number operator on the site i . Because the physical Hilbert space is generated by the electron operator which is an $SU(2)$ singlet, thus the $SU(2)$ charge on each site is always zero for the physical states

$$\left(\frac{1}{2}\psi_{a\alpha,i}^\dagger \psi_{b\alpha,i} + b_{a,i}^\dagger b_{b,i}\right)\tau_{ab}^l = 0, \quad l = 1, 2, 3 \quad (46)$$

The mean field Hamiltonian of (45) has a form

$$H_{mean} = \sum J_{ij} \text{Tr}(\psi_i^\dagger U_{ij} \psi_j + h.c.) + \sum \frac{1}{2} a_0^l \text{Tr}(\psi_i^\dagger \tau^l \psi_i) \\ + \sum \frac{t}{8} b_i^\dagger U_{ij} b_j + \sum a_0^l b_i^\dagger \tau^l b_i \quad (47)$$

If we take $b_2^\dagger b_2 = 0$ (45)–(47) will reduce to (29)–(31). It is obvious that (47) is manifestly $SU(2)$ invariant.

Now let us discuss the physical meaning of the gauge invariance. Let U_{ij} and U'_{ij} be two gauge equivalent mean field ansatz and $|\Phi\rangle$ and $|\Phi'\rangle$ be the two mean field states associated with the two ansatz. $|\Phi\rangle$ and $|\Phi'\rangle$ are related by a $SU(2)$ gauge transformation: $|\Phi\rangle = W|\Phi'\rangle$. The physical electron wave function associated with the mean field state is obtained by doing the Gutzwiller projection:

$$\Psi_e(\{z_i, \alpha_i\}) = \langle 0 | \prod_i c_{\alpha_i}(z_i) | \Phi \rangle \quad (48)$$

where c_α is given by (44) and (z_i, α_i) are the coordinate and the spin of the i -th electron. Since c_α is invariant under the gauge transformation $c_\alpha = W^{-1} c_\alpha W$, we can easily see that the two mean field states $|\Phi\rangle$ and $|\Phi'\rangle$ give rise to the same electron wave function if they are related by a gauge transformation. Thus the two gauge equivalent ansatz U_{ij} and U'_{ij} generate identical spin state.

As an application of the above $SU(2)$ formalism, we would like to demonstrate that the chiral spin state discussed in the last section actually has a $SU(2)$ gauge symmetry. To see this we first notice that the mean field order parameters in (10) and (16) give rise to the $SU(2)$ order parameter $U_{ij} \propto \pm i$. The $SU(2)$ gauge symmetry is unbroken because

U_{ij} is invariant under global $SU(2)$ gauge transformations. Including the $SU(2)$ gauge fluctuations the effective Lagrangian (26) now becomes

$$\mathcal{L} = \sum_{I=1,2} \left[-i\lambda_{I\alpha}^\dagger (\partial_t - ia_0^l \tau^l) \lambda_{I\alpha} + \frac{1}{2m_s} \lambda_{I\alpha}^\dagger (\partial_i - ia_i^l \tau^l)^2 \lambda_{I\alpha} \right] - \frac{1}{4\pi} a_\mu^l f_{\nu\lambda}^l \epsilon_{\mu\nu\lambda} + \dots \quad (49)$$

where $\lambda_{I\alpha} = (s_{I,\alpha}, \bar{s}_{I,-\alpha})$ is a $SU(2)$ doublet (see (36)) and $f_{\mu\nu}^l$ is the field strength of the $SU(2)$ gauge field. One can easily check that (49) reduce to (26) after setting $a_\nu^{1,2} = 0$. The Chern-Simons term in (49) describes a level 1 $SU(2)$ Chern-Simons topological theory. The spinons still have the $\theta = \frac{1}{2}\pi$ Abelian statistics despite the coupling to the $SU(2)$ gauge fields.²⁰ However since s and $\bar{s}$ are mixed by the $SU(2)$ gauge interactions, there are only two physical quasiparticles (with spin up and down) for each value of I . The quasiparticles created by s and $\bar{s}$ are gauge equivalent under the $SU(2)$ gauge transformations. More specifically the electron wave functions for a s spinon and a $\bar{s}$ spinon are given by

$$\Psi_s(\{z_i, \alpha_i\}) = \langle 0 | \prod_i c_{\alpha_i}(z_i) s_\alpha^\dagger | \Phi \rangle, \quad \Psi_{\bar{s}}(\{z_i, \alpha_i\}) = \langle 0 | \prod_i c_{\alpha_i}(z_i) s_\alpha | \Phi \rangle$$

Since the $SU(2)$ gauge transformation can rotation s into $s^\dagger$, one can show that the electron wave functions for the s spinon and the $\bar{s}$ spinon are the same. To obtain this result we have used the fact that the mean field chiral spin state $|\Phi\rangle$ is invariant under the global gauge transformations.

At finite doping the holons are described by (47) with U_{ij} given by the mean field ansatz (10) and (16). It easy to see that both $b_a|_{a=1,2}$ have the same spectrum (32) which contains two minimums. Thus in the continuum limit we have two species of holons described by

$$\mathcal{L} = \sum_{I=1,2} \left[-ib_{Ia}^\dagger (\partial_t - ieA_0 - ia_0^l \tau^l)_{ab} b_{Ib} + \frac{1}{2m_h} b_{Ia}^\dagger (\partial_i - ieA_i - ia_i^l \tau^l)_{ab}^2 b_{Ib} \right] - \frac{1}{4\pi} a_\mu^l f_{\nu\lambda}^l \epsilon_{\mu\nu\lambda} + \dots \quad (50)$$

(49) and (50) are the complete low energy effective theory governing the dynamics of the spinons and the holons. (49) and (50) are more correct than (26) and (33).

V. DYNAMICAL STABILITY OF THE MEAN FIELD THEORIES OF SPIN LIQUID STATES

Given the generalized mean field formalism, we would like to have a general discussion about the mean field theory of spin liquid state. At the “zeroth” order the density of the slave fermions can still fluctuates. Some times the slave fermion density fluctuations may even be gapless. In those cases the constraint (4) is badly violated. The zeroth order

mean field theory (40) may fail to provide even a qualitative description of the low energy properties of a true spin liquid state.

One may try to improve the zeroth order mean field theory by including the gauge field to enforce the constrain (4). The gauge interactions will make the density fluctuations of the slave fermions to have a finite energy gap. In this case the constraint (4) is satisfied at least in the infrared limit. But, a system of fermions with $SU(2)$ gauge interaction usually is so complicated that it is very difficult to obtain the low energy properties of the system. The difficulty comes from the gaplessness of the fermion excitations and the gauge excitations. The system has severe infrared divergence. The infrared properties of the system can hardly be determined. If this happens, the mean field theory essentially “maps” a complicated spin system into a (probably more) complicated fermion system with $SU(2)$ gauge interaction. The low energy properties of both systems are hard to obtain.

But, there are three possible cases in which the low energy properties of (43) can be relatively reliably determined.

A) The mean field solution U_{ij} breaks the translation symmetry and the slave fermions described by (40) (at the “zeroth” order) form a band insulator (Here we have assumed that $\eta_{ij} = 0$). In this case, even at the “zeroth” order (*i.e.*, without the $SU(2)$ gauge field fluctuations), the slave fermion gas is already incompressible. Due to the finite energy gap, we can safely integrate out the electrons and thus reducing (43) into a pure lattice gauge theory. The low energy properties of original spin system can be obtained from the low energy properties of the effective lattice gauge theory, which in many cases is still non-trivial. In general the gauge field mediate a confining interaction and the spin $\frac{1}{2}$ slave fermion cannot appear in physical spectrum.⁹ Only slave fermion pairs can appear as physical quasiparticles which carry integer spins. The spin excitations in general have finite gap. The spin-Peierls state discuss in Ref. 7 is an example of this type of spin states.

B) The mean field solution $U_{ij}^{(mean)}$ generates a flux (again assuming $\eta_{ij} = 0$). In this case the slave fermions described by (40) behave as if they are moving in a magnetic field. When the filling fraction is right, an integral number of Landau levels are completely filled. The slave fermion gas becomes incompressible due to the finite gap between Landau levels. Again the slave fermion density fluctuations are absent even at the “zeroth” order of the mean field theory. The time reversal symmetry and the parity are spontaneously broken in such a state. After integrating out the slave fermions the effective lattice gauge theory contains a Chern-Simons term. Because of the Chern-Simons term, the gauge field fluctuations have finite energy gap and can only mediate short range interactions. As a consequence the slave fermions are not confined. The quasiparticles (spinons) are dressed slave fermions which carry spin $\frac{1}{2}$ and have fractional statistics. The mean field chiral spin state studied in Ref. 11,10,12 is a typical example of this possibility.

C) The third way to obtain a reliable mean field theory is to break the $SU(2)$ gauge symmetry. In this case the gauge fluctuations obtain a finite energy gap due to the Anderson-Higgs mechanism which solve the infrared problem. The slave fermion gas is again incompressible since there is no gapless excitations in the Higgs phase.

Let us discuss the necessary conditions to break the $SU(2)$ gauge symmetry. The free energy of (43) is a function of $U_{ij}^{(mean)}$ and a_{ij}^l : $F(U_{ij}^{(mean)}, e^{ia_{ij}^l \tau^l})$. To discuss the properties of the free energy it is convenient to introduce the plaquette variable of the mean field solution

$$P_{ijk..li} = U_{ij}^{(mean)} U_{jk}^{(mean)} \dots U_{li}^{(mean)} \quad (51)$$

First let us assume all the plaquette variable P commute with a generator in the $SU(2)$ Lie algebra, say τ^3 . In this case one can choose a mean field ansatz (by performing gauge transformations) such that all $U_{ij}^{(mean)}$ have a form $ie^{i\phi_{ij}\tau^3}$. Now the gauge invariance of the free energy implies that

$$F(U_{ij}^{(mean)}, e^{ia_{ij}^l \tau^l}) = F(U_{ij}^{(mean)}, e^{\theta_i \tau^3} e^{ia_{ij}^l \tau^l} e^{-i\theta_j \tau^3}). \quad (52)$$

In order for the gauge bosons to obtain a finite Higgs mass we need all the three mass terms $(a^l)^2$ to appear in the expansion of the free energy. However one can check that the mass term $(a^3)^2$ is incompatible with (52). Therefore at least the gauge boson a^3 are massless if all P commute with τ^3 . Thus in order to completely breaks the $SU(2)$ symmetry we must require the plaquette variables at least take two different values which do not simultaneously commute with any $SU(2)$ generators. In this case η_{ij} must be non-zero. Once the $SU(2)$ gauge symmetry is broken, the infrared problems are well under control due to the finite mass term. In this case the low energy properties of the mean field theory and the corresponding spin liquid state can be reliably derived.

In all the cases discussed above the infrared problem of the slave fermions with the $SU(2)$ gauge interaction is resolved by opening energy gaps. Due to those energy gaps the infrared behavior of the mean field theory is well under control. We can obtain the low energy properties of the mean field theory quite reliably. Because of the infrared stability of the theory, those low energy properties are expected to be robust against small perturbations. Therefore it is reasonable to assume that the low energy properties of the mean field theory qualitatively describe the low energy properties of the spin liquid state. The mean field theories satisfying A), B) or C) are said to be dynamically stable. Many mean field theory studied before are not dynamically stable. It is thus dangerous to extract the low energy properties of the spin liquid state, *e.g.*, the quantum numbers of the quasiparticles, from those dynamically unstable mean field theories.

In the mean field theories discussed in A–C), the slave fermions form incompressible states. The dynamical stability of the mean field states and the incompressibility of the slave fermions are closely related. Given a slave fermion system one may ask what kind of the incompressible states that the slave fermions can form. There are only three well understood incompressible states for an fermion system. The fermions may form a band insulator which corresponds to the case A). If a “magnetic” field is dynamically generated the fermions may form an incompressible quantum Hall liquid which corresponds to the case B). The fermions can also form a superconducting state, which is again incompressible. Notice that a superconducting state, in contrast to a superfluid state, contains no gapless excitations. It certainly contains no gapless density fluctuations. This corresponds to the case C).

The spin liquid states for the case B) have been studied in detail in Ref. 11 and in section 3. In the next section, as an example, we will study a T and P symmetric spin liquid state corresponding to the case C).

VI. T AND P SYMMETRIC SPIN LIQUID STATE –

MEAN FIELD THEORY OF THE S-RVB STATE.

The T and P symmetric spin liquid state²¹ is given by the following mean field ansatz:

$$\begin{cases} \chi_{i, i+\hat{x}} = \chi_{i, i+\hat{y}} = \chi \\ \eta_{i, i+\hat{x}+\hat{y}} = \eta_{i, i-\hat{x}+\hat{y}}^\dagger = \eta \\ \chi_{ij} = \eta_{ij} = 0, \text{ otherwise,} \\ a_0^1 \neq 0, \quad a_0^2 = a_0^3 = 0 \end{cases} \quad (53)$$

where χ is a real parameter and η is a complex parameter. The corresponding $SU(2)$ link variables U_{ij} are given by:

$$\begin{aligned} U_{i, i+\hat{x}} &= U_{i, i+\hat{y}} = -\chi\tau^3 \\ U_{i, i+\hat{x}+\hat{y}} &= \text{Re}(\eta)\tau^1 + \text{Im}(\eta)\tau^2 \\ U_{i, i-\hat{x}+\hat{y}} &= \text{Re}(\eta)\tau^1 - \text{Im}(\eta)\tau^2 \end{aligned} \quad (54)$$

Later we will show that the constraint (5) can be completely satisfied by properly choosing the value of a_0^1 .

From the mean field ansatz we can easily obtain the following results:

1) The mean field ansatz is invariant under the translation and the spin rotation. Therefore, the mean field ansatz describes a spin liquid state. Or more precisely the ground state of H in (40), after the Gutzwiller projection, gives rise to a wave function of spin liquid state which respects the translation symmetry.

2) $U_{ij} = U_{ji}^\dagger = U_{ji}$. Thus the links are non-oriented.

3) Around triangles we have

$$\begin{aligned} U_{23} U_{31} U_{12} &= U_{43} U_{31} U_{14} = -\chi^2(\text{Re}(\eta)\tau^1 + \text{Im}(\eta)\tau^2) \\ U_{12} U_{24} U_{41} &= U_{32} U_{24} U_{43} = -\chi^2(\text{Re}(\eta)\tau^1 - \text{Im}(\eta)\tau^2) \end{aligned} \quad (55)$$

where 1,2,3,4 label the four corners of a plaquette anti-clockwise. Because τ^1 and τ^2 do not commute, the $SU(2)$ gauge symmetry is completely broken if $\text{Re}\eta, \text{Im}\eta \neq 0$. All gauge bosons obtain finite masses (or, energy gaps).²²

4) The mean field ansatz is invariant under the parity $P : x \leftrightarrow y$. Therefore the spin liquid state is P invariant.

However, the mean field ansatz is not invariant under 90° rotation and one might naively expect that the spin liquid state breaks the 90° rotation symmetry. But remember that U_{ij} are not physically observable. Only the $SU(2)$ gauge equivalent classes of U_{ij} are physical variables. Therefore the non-invariance of U_{ij} under the 90° rotation does not imply the non-invariance of the gauge equivalence classes of U_{ij} . In the following we will

show that the gauge equivalent classes of the mean field ansatz (53) (or (54)) is invariant under the 90° rotation. Hence the corresponding spin liquid state respects the 90° rotation symmetry.

Under the 90° rotation U_{ij} transform into U'_{ij} where

$$\begin{aligned} U'_{i, i+\hat{x}} &= U'_{i, i+\hat{y}} = -\chi\tau^3 \\ U'_{i+\hat{x}+\hat{y}} &= \text{Re}(\eta)\tau^1 - \text{Im}(\eta)\tau^2 \\ U'_{i-\hat{x}+\hat{y}} &= \text{Re}(\eta)\tau^1 + \text{Im}(\eta)\tau^2 \end{aligned} \quad (56)$$

The two mean field ansatz U_{ij} and U'_{ij} can be shown to be gauge equivalent

$$U_{ij} = W_i U'_{ij} W_j^\dagger \quad (57)$$

where

$$W_i = (-)^i \tau^1 \quad (58)$$

The gauge transformation W_i also leave a_0^1 invariant. From the discuss in the last section we see that U_{ij} and U'_{ij} give rise to the same electron wave function after the Gutzwiller projection. Therefore the mean field ansatz (16) actually describe a spin liquid state which is invariant under the 90° rotation.

Under T the mean field ansatz U_{ij} is changed to $U'_{ij} = U_{ij}^*$. Using the same gauge transformation (58) we can show that U_{ij} and U'_{ij} are gauge equivalent. Thus our spin liquid state also respects time reversal symmetry.

Let us discuss the point 3 in more detail and identify the terms that can generate the Higgs mass for gauge fields. First we would like to make a gauge transformation $W_i = (i\tau^3)^i$ to transform the ansatz (54) into

$$\begin{aligned} U_{i, i+\hat{x}} &= U_{i, i+\hat{y}} = i\chi \\ U_{i, i+\hat{x}+\hat{y}} &= (-)^i (\text{Re}(\eta)\tau^1 + \text{Im}(\eta)\tau^2) \\ U_{i, i-\hat{x}+\hat{y}} &= (-)^i (\text{Re}(\eta)\tau^1 - \text{Im}(\eta)\tau^2) \end{aligned} \quad (59)$$

Notice that in the new ansatz $U_{i, i+\hat{x}} = U_{i, i+\hat{y}} \propto 1$. The plaquette variables in (55) now become

$$\begin{aligned} P_A(i) &= -(-)^i \chi^2 (\text{Re}(\eta)\tau^1 + \text{Im}(\eta)\tau^2) \\ P_B(i) &= -(-)^i \chi^2 (\text{Re}(\eta)\tau^1 - \text{Im}(\eta)\tau^2) \end{aligned} \quad (60)$$

In general the free energy contains, for example, the following gauge invariant terms

$$F = a\text{Tr}[P_A(i)U_{i,i+\hat{x}}P_A(i+\hat{x})U_{i+\hat{x},i}] + b\text{Tr}[P_B(i)U_{i,i+\hat{x}}P_B(i+\hat{x})U_{i+\hat{x},i}] + \dots \quad (61)$$

For small gauge fluctuations $U_{i,i+\hat{x}} = i\chi e^{ia_x^l \tau^l}$ and (61) can be written as

$$F = -\frac{1}{2}a\chi^2\text{Tr}([P_A, a_x^l \tau^l]^2) - \frac{1}{2}b\chi^2\text{Tr}([P_B, a_x^l \tau^l]^2) + \dots \quad (62)$$

We see from (62) that the mass term for all three $SU(2)$ gauge fields $a^l|_{l=1,2,3}$ are generated if both $\text{Re}(\eta)$ and $\text{Im}(\eta)$ are non-zero.

Let $|\Phi\rangle$ be the ground state of the mean field Hamiltonian (40). We would like to view $|\Phi\rangle$ as a trial wave function. The mean field ground state energy is given by $E = \langle \Phi | H | \Phi \rangle$ where H is given by (3). Notice that for the mean field ansatz (53) the mean field Hamiltonian H_{mean} can be written as

$$\begin{aligned} \frac{H_{mean}}{J_1\chi} = & \sum_{n.n.} s_{i\alpha}^\dagger s_{j\alpha} + \sum_{n.n.n.} (\zeta s_{i\alpha} s_{j\beta} \epsilon^{\alpha\beta} + h.c.) \\ & + \sum (\tilde{a}_0^1 s_{i\alpha} s_{i\beta} \epsilon_{\alpha\beta} + h.c.) \end{aligned} \quad (63)$$

where $\zeta = \frac{J_2\eta}{J_1\chi}$ and $\tilde{a}_0^1 = a_0^1/J_1\chi$. From (63) we see that $|\Phi\rangle$ only depend on ζ ($\tilde{a}_0^1$ is determined by (5)). Thus the mean field energy E is a function of ζ .

When $J_2/J_1 = 0$ we find that $E(\zeta)$ is minimized at $\zeta = 0$. The mean field solution is given by $\chi = 0.41$ and $\eta = 0$. When $0 < J_2/J_1 < 1.4$, $E(\zeta)$ is minimized at a non-zero ζ with $\text{Im}\zeta, \text{Re}\zeta \neq 0$. Hence the mean field solution also has $\text{Im}\eta, \text{Re}\eta \neq 0$ and the $SU(2)$ gauge symmetry is completely broken. If J_2/J_1 is not too small (*e.g.*, $J_2/J_1 > 0.3$) we have $\text{Im}\eta/\text{Re}\eta \sim 1$. When $J_2/J_1 > 1.4$ we find that $E(\zeta)$ is minimized at $|\zeta| = \infty$ and $\text{Im}\zeta/\text{Re}\zeta = 1$. In this case the mean field solution is given by $\chi = 0$ and $\eta = 0.48e^{i\pi/4}$.

We would like to remark that although the mean field ansatz with $\text{Im}\eta, \text{Re}\eta \neq 0$ minimize the mean field ground state energy for $0 < J_2/J_1 < 1.4$, this does not imply that the spin Hamiltonian (1) support such a spin liquid state. This is because the quantum fluctuations are large and the mean field calculation is not reliable. The above calculation is just for illustrative purpose. In the following we will assume that the mean field solution given by (53) with $\text{Im}\eta, \text{Re}\eta \neq 0$ is stable. In this case the $SU(2)$ gauge symmetry is completely broken. If the simplest Heisenberg model does not give rise to this result, we will assume such a result can be realized in some other spin models.

Assuming the stability of the mean field ground state, we can study the properties of the quasiparticle excitations. At the “zeroth” order, the dynamics of the quasiparticles is described by the mean field Hamiltonian (40). Notice that the mean field ansatz actually describes a BCS “superconductor”. We may directly apply the results of the BCS theory to our case. We would like to point out that the “superconducting” order parameter in our case has a $s + id$ symmetry. The order parameter does not have a definite angular momentum. This is possible because the gap equation is non-linear. From (40) we see that “superconducting” gap is given by

$$\Delta_k = 2J_2 \left[\eta \cos(k_x + k_y) + \eta^\dagger \cos(-k_x + k_y) \right] + a_0^1 + ia_0^2 \quad (64)$$

and the slave fermion spectrum is given by

$$\epsilon_k = 4J_1\chi(\cos k_x + \cos k_y) + a_0^3 \quad (65)$$

a_0^l are Lagrangian multipliers which are chosen enforce the constraint

$$\chi_{ii} = 1, \quad \eta_{ii} = 0 \quad (66)$$

χ_{ij} and η_{ij} can be determined from Δ_k and ϵ_k :

$$\begin{aligned}\chi_k &= \sum_j \chi_{ij} e^{i\mathbf{k}\cdot(\mathbf{j}-\mathbf{i})} = \left(1 - \frac{\epsilon_k}{E_k}\right) \\ \eta_k &= \sum_j \eta_{ij} e^{i\mathbf{k}\cdot(\mathbf{j}-\mathbf{i})} = \frac{1}{2} \frac{\Delta_k}{E_k}\end{aligned}\tag{67}$$

where

$$E_k = \sqrt{\epsilon_k^2 + |\Delta_k|^2}\tag{68}$$

The constraint (66) reduces to

$$\begin{aligned}\chi_{ii} &= \frac{1}{N} \sum_k \left(1 - \frac{\epsilon_k}{E_k}\right) = 1 \\ \eta_{ii} &= \frac{1}{N} \sum_k \frac{1}{2} \frac{\Delta_k}{E_k} = 0\end{aligned}\tag{69}$$

We find that the constraint (69) can be satisfied if we choose $a_0^3 = a_0^2 = 0$. But in general $a_0^1 \neq 0$.

From the BCS theory, we see that the spectrum of the quasi-particle excitations is given by E_k . The quasi-particles are fermions with $\frac{1}{2}$ spin and zero electrical charge. Notice that the quasi-particle excitations has finite energy gap, as one can see from (68).

We would like to stress that the mean field state studied above is *not* an (electrical) superconductor. It is only a “superconductor” for the dynamical (sometimes is called fictitious) gauge field a_μ^3 arisen from the phase fluctuations of χ_{ij} . For the electromagnetic field our mean field state is an insulator.

In the “first” order mean field theory the collective fluctuations of the phases of U_{ij} are included. From the previous discussion we see that these fluctuations correspond to an $SU(2)$ gauge field. In the mean field state given by (53), the $SU(2)$ gauge symmetry is broken. The gauge bosons obtain finite masses. The infrared behavior of the $SU(2)$ gauge fluctuations are well under control. Those fluctuations do not qualitatively change the structure of the “zeroth” order mean field ground state. The massive $SU(2)$ gauge fluctuations can only mediate short range interactions between quasi-particles in the “zeroth” order mean field theory, and there is no confinement. Therefore the quasi-particles in the “zeroth” order mean field theory also appears as the quasi-particles in the “first” order mean field theory. Those quasi-particles qualitatively describe the excitations in the corresponding spin liquid state. In particular the spin liquid state considered here support neutral spin $\frac{1}{2}$ fermionic excitations. These excitations are spinons. The energy spectrum of the spinons is qualitatively given by E_k and has finite energy gap.

To summarize we have described a mean field theory of a spin liquid state. The spin liquid state respects the translation symmetry and the 90° rotation symmetry. It also respects T and P . The spin excitations in the spin liquid state are found to be neutral spin $\frac{1}{2}$ spinons. The spinons obey Fermi statistics and have a finite energy gap. Such a spin liquid correspond to the s-RVB states.²³ We would like to emphasize that the mean field theory discussed above is dynamically stable. Therefore, we expect the mean field

theory qualitatively describes the properties of the actual spin liquid state supported by some spin Hamiltonian.

Now let us study the doped s-RVB states in the low doping limit. In presence of both order parameters χ_{ij} and η_{ij} , the mean field Hamiltonian of the t - J model at non-zero doping is given by (45). In our case it reduces to

$$H_{mean} = \sum J_{ij} \left[\chi_{ij} s_{i\alpha}^\dagger s_{j\alpha} + \left(\eta_{ij} s_{i\alpha}^\dagger s_{j\beta}^\dagger \epsilon^{\alpha\beta} + h.c. \right) \right] + \sum t \chi_{ij} (b_{1j}^\dagger b_{1i} - b_{2j}^\dagger b_{2i}) + \sum a_0^1 (s_{i\alpha} s_{i\beta} \epsilon_{\alpha\beta} + b_{1i}^\dagger b_{2i} + b_{2i}^\dagger b_{1i}) \quad (70)$$

In (70) b describes a charge e spinless boson which is called holon. At zeroth order, the holons have a dispersion:

$$E_{k,\pm}^h = \pm \sqrt{(4t\chi)^2 (\cos k_x + \cos k_y)^2 + (a_0^1)^2} \quad (71)$$

Because χ is of order $O(1)$, the effective mass of the holon is of order $\frac{1}{t}$ which is close to the electron band mass. But in (71) we have ignore the fluctuations of χ_{ij} and η_{ij} . The actual holons are dressed by those fluctuations and has much larger effective mass. We also see that $E_{k,-}^h$ has two minimums at $(0,0)$ and $(\frac{\pi}{a}, \frac{\pi}{a})$. Therefore there are two species of holons in the continuum limit.

The phenomenological implications of the spinon-holon model (70) have been studied by many people based on the mean field theory and on an assumption that the interaction between the spinons and the holons are weak.²⁴ However, many mean field theories studied before contain strong gauge field fluctuations which mediate a confining interaction between the spinons and the holons. Here we find a mean field spin liquid state in which the $SU(2)$ gauge field are massive and the interaction between the spinons and the holons is relatively weak and is short ranged. Thus many results of the spinon-holon model obtained in the previous studies may apply to our spin liquid state.

In addition to the spinons and the holons, there is another type of quasi-particle excitations in our spin liquid state. This excitation appears as topological soliton in the mean field theory. Notice that in our mean field state the $SU(2)$ gauge symmetry is broken by the Higgs fields in the adjoint representation of the $SU(2)$. In this case the $SU(2)$ gauge symmetry is broken down to Z_2 gauge symmetry. The quasi-particles in the Z_2 gauge theory are the Z_2 -vortices. These Z_2 -vortices are the new quasi-particle excitations mentioned above. In the mean field theory a Z_2 -vortex is described by the following ansatz:

$$\tilde{U}_{ij} = e^{-i\theta_i \tau^3/2} U_{ij}^{(mean)} e^{i\theta_i \tau^3/2} \quad (72)$$

where θ_i is the angle of the i -th site relative to the center of the Z_2 -vortex (x, y) :

$$\text{tg } \theta_i = \frac{i_x - x}{i_y - y} \quad (73)$$

Because $e^{i\theta \tau^3/2}|_{\theta=2\pi} = -1$, the holon (spinon) wave function will obtain a minus sign as a holon (spinon) moves around the Z_2 -vortex. Therefore the Z_2 -vortex behaves like a π -flux vortex to the spinons and the holons.

Because of the Z_2 -vortex, the $\frac{hc}{2e}$ flux have a finite energy even in the charge e holon condensed state.²⁵ In the superconducting state a Z_2 -vortex and a bare $\frac{hc}{2e}$ magnetic vortex have infinite energy since the holon wave function change sign as a holon goes around the Z_2 -vortex or the $\frac{hc}{2e}$ magnetic vortex. But the bound state of a Z_2 -vortex and a $\frac{hc}{2e}$ magnetic vortex has a finite energy. The holon wave function does not change sign as a holon moves around such a bound state. Therefore even in the charge e holon condensed state, the minimum flux quantum is still $\frac{hc}{2e}$. The charge e holon superconducting state does not contradict with the experimental results. We also notice that binding a Z_2 -vortex to a holon change the statistics of the holon from bosonic to fermionic. Similarly, the bound state of a spinon and a Z_2 -vortex behave like a boson. This phenomenon has been discussed in detail in Ref. 26.

VII. TOPOLOGICAL ORDERS IN THE SPIN LIQUID STATES

In this article and in Ref. 11 the s-RVB state and the chiral spin state are constructed by using the mean field theory. Both spin liquid states have finite energy gap and respect the translation symmetry. At half filling such spin liquid states are Mott insulators, or more precisely, an insulator with *odd* number of electrons per unit cell.

However, the spin liquid states are constructed by using the mean field theory which involves unphysical fields (*e.g.*, the slave fermion field s_i and slave boson field b_i). Our description and characterization of the spin liquid states are also in term of those unphysical quantities (like the mean field “order” parameters χ_{ij} and η_{ij}). It is very important to understand how to characterize the spin liquid states (or the Mott insulators) by using physical properties. Especially we would like to know whether there are any physical order parameters which characterizes the spin liquid phase studied in this article.

In this section we are going to argue that the spin liquid states studied here contain new types of orderings which cannot be characterized by the symmetries and by the order parameters associated with broken symmetries. This type of orderings is called topological orders. The spin liquid states discussed in section 3 and section 6 contain non-trivial topological orders and are characterized by those topological orders.

First let us discuss what is the topological orders.²⁷ Consider a rigid state containing no gapless quasiparticle excitations. Because of the finite energy gap such a system is almost trivial at low energies. The only non-trivial feature at low energies comes from degenerate ground states. One may naively expect the ground state degeneracy to come from broken symmetries and conclude that the rigid states are classified by their symmetry properties. However this naive expectation is not always correct. It has been shown that the fractional quantum Hall (FQH) liquids support degenerate ground states even when the Hamiltonian contains no discrete symmetries.²⁸ The ground state degeneracy is shown to be robust against arbitrary perturbations despite there is no symmetry to protect it. Furthermore the number of the degenerate ground states depend on the topology of the space. All of these results point to one thing: rigid systems, such as FQH states, may have non-trivial infrared fixed points which cannot be characterized by broken symmetries. Those unusual infrared fixed points indicate the existence of new kind of orderings which are called the topological orders.

To give a physical definition of the topological orders we need to specify physically measurable quantities which characterize the topological orders. The ground state degen-

eracy discussed above is one such quantity. Certainly the ground state degeneracy alone cannot completely characterize the topological orders. More complete characterizations can be obtained through the non-Abelian Berry's phase²⁷ and the edge excitations.²⁹

Both the chiral spin state and the s-RVB state are rigid states. In Ref. 30,27 the chiral spin states are shown to have non-trivial topological orders. We will review this result later. Right now we would like to show that the s-RVB state also contains non-trivial topological orders.

As in the FQH states, the non-trivial topological orders in the s-RVB state can be detected by measuring the ground state degeneracy of the system on compactified spaces. It can be shown that our s-RVB state has 2^{2g} degenerate ground states on a genus g Riemann surfaces. In the mean field theory the 2^{2g} degenerate ground states can be constructed by adding zero or one unit of the Z_2 -flux through the $2g$ non-contractible loops on the genus g Riemann surface. On the torus the mean field ansatz for the four degenerate ground states is given by:

$$U_{ij}^{(m, n)} = e^{-i(m\frac{ix}{L_x} + n\frac{iy}{L_y})\tau^3\pi} U_{ij}^{(mean)} e^{i(m\frac{ix}{L_x} + n\frac{iy}{L_y})\tau^3\pi} \quad (74)$$

where $m, n = 0, 1$ and L_x and L_y are the size of torus in the x and y directions. $U_{ij}^{(m, n)}$ with different m and n are *locally* gauge equivalent. Therefore, the free energies for different ansatz are expected to be the same (in the thermodynamical limit). Thus $U_{ij}^{(m, n)}$ describes the degenerate ground states of the system. However $U_{ij}^{(m, n)}$ with different m and n are not gauge equivalent in the global sense. A spinon propagating all the way around the torus in the $x(y)$ direction obtain a phase $e^{im\pi}$ ($e^{in\pi}$). Therefore $U_{ij}^{(mn)}|_{m, n=0,1}$ describes different orthogonal ground states. We would like to mention that the mean field ansatz with m (or n) differ by an even integer are gauge equivalent and give rise to the same physical spin wave function.

On a finite compactified lattice, the only way for the system to tunnel from one ground state to another is through the following tunneling process. At first a pair of the Z_2 -vortices is created. One of the Z_2 -vortices propagates all the way around the torus and then annihilates with the other Z_2 -vortex. Such a process effectively add a unit of the Z_2 flux to the hole of the torus and change m or n by 1. The different ground states can not tunnel into each other through any local fluctuations because of the flux conservation. As a direct consequence of this result, the energy split between different ground state on a finite lattice is expected to be of order $e^{-L/\xi}$.

In the mean field theory the degeneracy of ground states is a consequence of the gauge symmetry. The gauge symmetry remains to be exact even after we include an arbitrary perturbation to the original spin Hamiltonian:

$$\delta H = \sum \delta J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \dots \quad (75)$$

δH may break translation symmetry, rotation symmetry, *etc.* . The above arguments are still valid even for the modified Hamiltonian. The mean field ground states remain to be four fold degenerate. We expect this result to be valid even beyond the mean field theory. The ground state degeneracy of the s-RVB state can not be changed by any perturbations as long as the perturbations are weak enough not to drive a phase transition. Therefore,

the ground state degeneracy can be regarded as a quantum number characterizing the spin liquid state.

From the above discussions we can see that the ground state degeneracy of the s-RVB state is not due to broken discrete symmetries. This is because 1) The ground state degeneracy depends on the topology of the compactified space; 2) the ground state degeneracy is robust against arbitrary perturbations, even those perturbations which break all the symmetries in the Hamiltonian. 3) The energy split of the ground states is of order $e^{-L/\xi}$ for a finite system of size L . If the ground state degeneracy was due to broken discrete symmetry the energy split would be at most of order E^{-L^2/ξ^2} . This result suggest that the s-RVB state studied in section 6 contains non-trivial topological orders.

The topological order is a very useful concept. Let us consider the following question: what is the difference between the spin-Peierls state and the s-RVB state? One may immediately say that the two states have different symmetries. But if we modify our Hamiltonians to break the translation and the rotation symmetries, then the two states will have the same symmetries. In this case we can still ask whether the two states are the same or not in the sense whether one state can be continuously deform into the other without phase transitions. When the translation and the rotation symmetries are broken, the spin-Peierls state only support a non degenerate ground state. While the s-RVB state still has four degenerate ground states on the torus. Therefore the spin-Peierls state and the s-RVB state are different even when they have the same symmetries. The two states differ by having different topological orders.

Now let us consider the topological orders in the chiral spin state. Again we would like to calculate the ground state degeneracy of the chiral spin state on, say, a torus. The easiest way to calculate the ground state degeneracy is to use the effective theory (22). The degenerate ground states arise from the following collective fluctuations

$$a_i(t, x, y) = \theta_i(t)/L_i, \quad i = x, y \quad (76)$$

All other fluctuations generate non zero “magnetic” field $b = f_{xy}$ and have finite energy gap as one can see from the classical equation of motion. The Lagrangian describing the dynamics of the collective excitations in (76) can be obtained by plugging (76) into (22)

$$L = \frac{2}{4\pi}(\dot{\theta}_x\theta_y - \dot{\theta}_y\theta_x) + \frac{1}{4\pi g^2}\dot{\theta}_i^2. \quad (77)$$

Notice that under the gauge transformation $U(x, y) = e^{2\pi i(\frac{nx}{L_x} + \frac{my}{L_y})}$ the gauge field (a_x, a_y) transform into a'_i ($a'_i = a_i - iU^{-1}\partial_i U$):

$$(a'_x, a'_y) = (a_x + \frac{2\pi n}{L_x}, a_y + \frac{2\pi m}{L_y}) \quad (78)$$

where n, m are integers. Note U is single valued on the torus when and only when m and n are integers. (78) implies that (θ_x, θ_y) and $(\theta_x + 2\pi n, \theta_y + 2\pi m)$ are gauge equivalent and should be identified. The gauge inequivalent configurations are given by a point on a torus $0 < \theta_i < 2\pi$.

The Lagrangian (77) describes a particle with unit charge moving on a torus parameterized by (θ_x, θ_y) . The first term in (77) indicates that there is a uniform “magnetic”

field $B = 2/2\pi$ on the torus. The total flux passing through the torus is equal to $2\pi \times 2$. The Hamiltonian of (77) is given by

$$H = \frac{1}{4\pi g^2} \left[-(\partial_{\theta_x} - iA_{\theta_x})^2 - (\partial_{\theta_y} - iA_{\theta_y})^2 \right]. \quad (79)$$

The energy eigenstates of (79) form Landau levels. The gap between Landau levels is $\frac{1}{4\pi^2 g^2}$ which is independent of the size of the system. The number of the states in the first Landau levels is equal to the number of the flux quanta passing through the torus which is 2 in our case. Thus the ground state degeneracy is 2.

In the above we only discuss the ground states in one sector of the T and P breaking ground states, say the sector with $\langle \mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3) \rangle > 0$. The other sector with $\langle \mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3) \rangle < 0$ also give rise to two degenerate ground states. Thus the total ground state degeneracy is $4 = 2 \times 2$. A factor 2 comes from the T and P breaking. The other factor 2 comes from the collective excitations discussed above. The four ground states carry the following quantum numbers of the parity P and the 90° rotation R_{90} :²⁷

$$(P, R_{90}) = (+, +), (+, -), (-, +), (-, -) \quad (80)$$

The two fold ground degeneracy arisen from the collective excitations is due the non-trivial topological orders in the chiral spin state. This degeneracy depend on the the topology of the space. On a genus g Riemann surface the collective excitations give rise to 2^g degenerate ground states. This degeneracy again can be shown to be robust against arbitrary perturbations. We would like to mention that the ground state degeneracy of the chiral spin state is close related the fractional statistics of the spinons. In general one can show that if the spinons have a fractional statistics $\theta = \frac{p\pi}{q}$ the chiral spin state will have q^g fold degenerate ground states on a genus g Riemann surface.²⁸

Before ending this section we would like to mention that the FQH states also contain non-trivial topological orders. The topological orders in a large class of the FQH states are classified by integer valued symmetric matrices K (up to an equivalence relation $K \sim O^T K O$ where $O \in SL(N, Z)$ is a integer valued matrices of unit determinant).³¹ All the quasiparticle quantum numbers (like statistics and charge) can be obtained from the matrix K . The hierarchy states constructed by Haldane and Halperin were shown to be characterized by tri-diagonal matrices.

Due to close relation between the chiral spin states and the FQH states, one can also construct more general chiral spin states by using the similar construction in the FQH theory. The above results for the FQH states apply equally well to the chiral spin states. In other words the FQH states and chiral spin states contain the same types of the topological orders.

VIII. SUMMARY

In this article we studied some mean field theories of spin liquid states. In those mean field theories the density fluctuations have finite gap. Because the system is already incompressible at mean field level, the mean field theories are able to give a correct qualitative

description of the spin liquid states. Many properties of the spin liquid states can be reliably obtained from the mean field theories. In particular for the cases B) and C) discussed in section 5, the gauge fluctuations in the mean field theories have finite gap and do not mediate any confining interactions. In those cases, the quasi-particle in the mean field theories directly correspond to the quasi-particles in the spin liquid states.

In the chiral spin state the gauge bosons are massive due to the Chern-Simons term. The spinons and the holons in the chiral spin state carry fractional statistics. Since an anyon gas automatically becomes superconducting at low temperatures, the chiral spin model has a potential to explain the high T_c superconductivity. We also studied another dynamically stable mean field spin liquid state which respect the T and P . The gauge bosons obtain finite masses from the Anderson-Higgs mechanism. The quasi-particles in the corresponding spin liquid state are found to be the spinons with Fermi statistics and the holons with Bose statistics. Such a state corresponds to the s-RVB state. The s-RVB model is also a potential theory for the high T_c superconductivity (at least in the low doping limit).

It is demonstrated that the spin liquid states of strongly correlated systems in general cannot be characterized by order parameters associated with broken symmetries. The spin liquid states may contain a new type of internal orderings which is called the topological orders. The definition (or the physical measurement) of the topological orders was discussed. Both the s-RVB state and the chiral spin states are shown to contain non-trivial topological orders. The existence of the topological orders is expected to be a general phenomenon for strongly correlated states with finite gaps. The results obtained in sections 6 and 7 are not limited in two dimensions. They apply equally well to higher dimensions.

We would like to mention that the s-RVB states and the chiral spin states represent new kinds of universality classes of insulators. Namely the insulators with odd number of electrons per unit cell. Those new universality classes of insulators are characterized by topological orders. The holes in the new insulators have unusual properties as we have seen from the discussions in this article. Thus the doped insulators become some sort of strange metals (*e.g.*, boson metals or semion metals). It would be very interesting to see whether those strange metals can explain the unusual normal state properties observed in the high T_c samples.

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